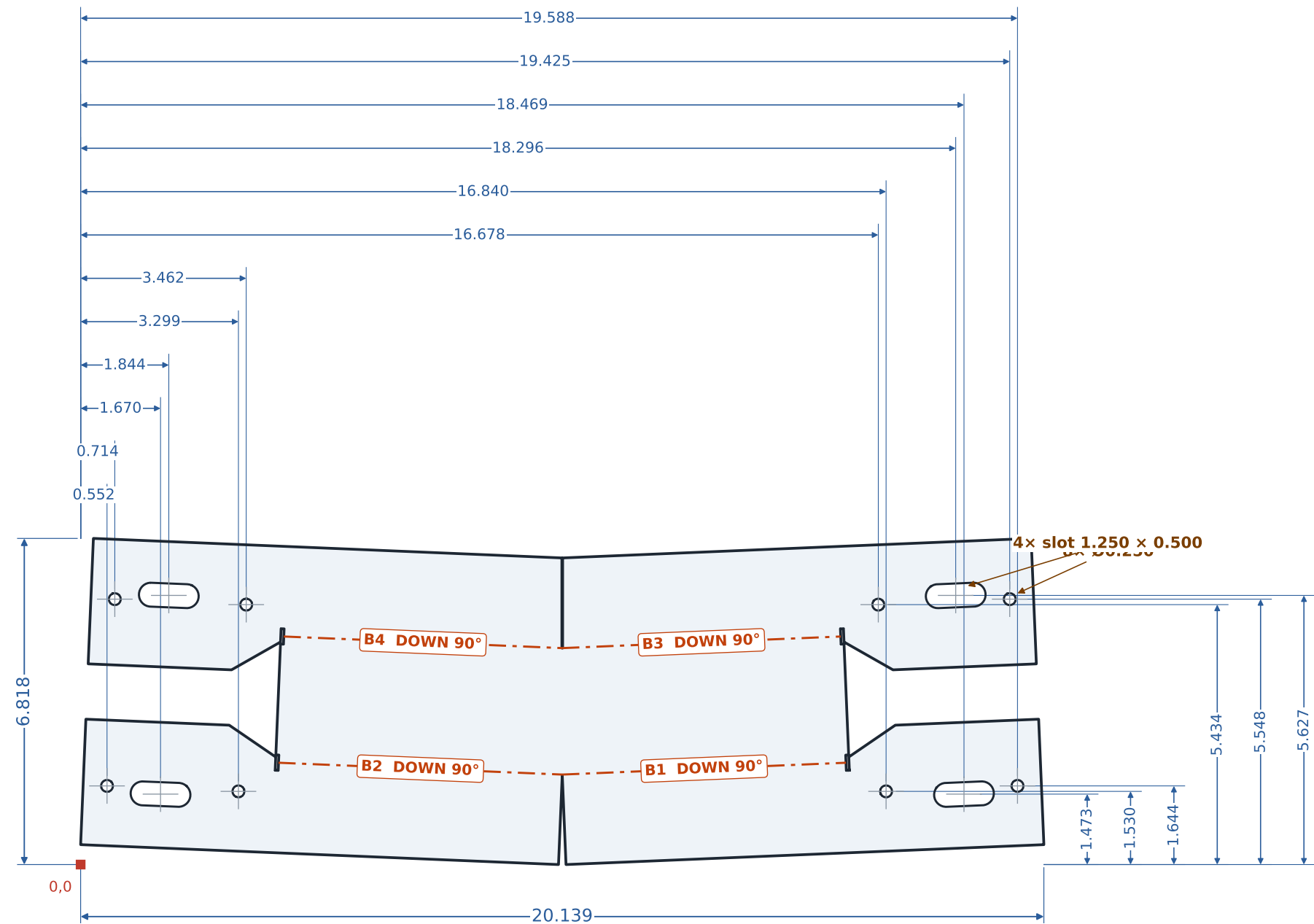


FF-1 · Channel with ears — flat pattern, one piece (4 bends 90°, web on top)

10 ga 0.135" (3.42 mm) · HR steel sheet A1011 CS · flat blank 20.139" × 6.818" · ≈ 4.35 lb each · QTY 1 per vehicle

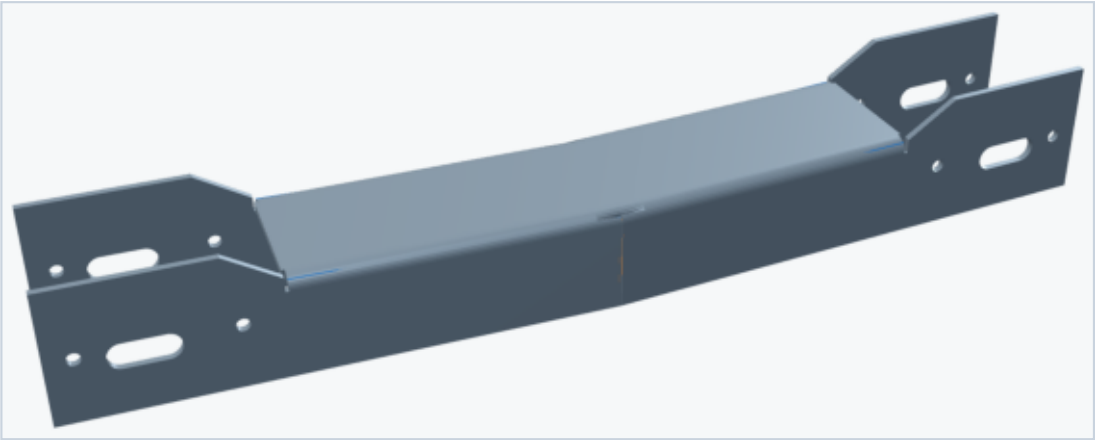
FLAT PATTERN (as laser-cut, marked side up)



NOTES

- 1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
 - 2. Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
 - 3. Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
 - 4. Deburr all edges, remove dross.
- * one blank: the web is a chevron 6"+6" (ends ¼" out); bend the walls 90° down on the dashed lines, each in 2 hits (6" to the middle); outer wall has a V-notch — closes after bending, weld it; inner wall has a slit — opens ~0.16" after bending, weld it; the ears reach past the bend line; 1/16" relief slit at the bend end

FORMED PART (after bending)



BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	90.0°	DOWN (away)	R=t 0.135	1	(10.070, 1.881)→(16.002, 2.129)
B2	90.0°	DOWN (away)	R=t 0.135	2	(4.137, 2.129)→(10.070, 1.881)
B3	90.0°	DOWN (away)	R=t 0.135	3	(10.070, 4.525)→(15.892, 4.768)
B4	90.0°	DOWN (away)	R=t 0.135	4	(4.247, 4.768)→(10.070, 4.525)

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	Ø0.250	19.588	1.644	laser
H2	Ø0.250	19.425	5.548	laser
H3	slot 1.250×0.500	18.469	1.473	laser
H4	slot 1.250×0.500	18.296	5.627	laser
H5	Ø0.250	16.840	1.530	laser
H6	Ø0.250	16.678	5.434	laser
H7	Ø0.250	3.462	5.434	laser
H8	Ø0.250	3.299	1.530	laser
H9	slot 1.250×0.500	1.844	5.627	laser
H10	slot 1.250×0.500	1.670	1.473	laser
H11	Ø0.250	0.714	5.548	laser
H12	Ø0.250	0.552	1.644	laser

FILES

- FF-1_cut.dxf (qty 1 / vehicle)
- FF-1_ref.dxf (reference: + bend lines, drill-later)

FF-1	Front channel with ears (chevron 6"+6")	HR steel sheet A1011 CS 10 ga 0.135" (3.42 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	REV A
FF-1 · Channel with ears — flat pattern, one piece (4 bends 90°, web on top)		QTY 1 per vehicle	Tol.: cut ±0.010" · bend ±1° · IR = t	2026-10-03	SHEET 1/1